

SEONGLAE CHO

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PUBLICATIONS

- **Cho, S., Wu, Z., & Koshiyama, A. (2025).** *CorrSteer: Generation-Time LLM Steering via Correlated Sparse Autoencoder Features*. **ICML 2026 main**. <https://arxiv.org/pdf/2508.12535>
- **Cho, S., Wu, Z., & Koshiyama, A. (2026).** *Control Reinforcement Learning: Interpretable Token-Level Steering of LLMs via Sparse Autoencoder Features*. **ICLR 2026** Principled Design for Trustworthy AI. <https://arxiv.org/abs/2602.10437>
- **Cho, S., Wu, Z. et al. (2026).** *The Confidence Manifold: Geometric Structure of Correctness Representations in Language Models*. **ICLR 2026** Agentic AI in the Wild: From Hallucinations to Reliable Autonomy. <https://arxiv.org/abs/2602.08159>
- **Cho, S., Wu, Z. et al. (2026).** *Automata from Agent Traces: Failure and Next-Step Prediction*. **EMNLP 2026** under review.
- **Cho, S., Wu, Z. et al. (2026).** *Are Single-Token Sparse Autoencoder Features Irreplaceable? The Impact of Training Methodology*. **EMNLP 2026** under review (ACL ARR, March 2026).
- **Cho, S. et al. (2025).** *FaithfulSAE: Towards Capturing Faithful Features with Sparse Autoencoders without External Datasets Dependency*. In Proceedings of the **ACL 2025 SRW**. <https://aclanthology.org/2025.acl-srw.20/>
- **Wu, Z., Cho, S. et al. (2025).** *AgentGraph: Trace-to-Graph Platform for Interactive Analysis and Robustness Testing in Agentic AI Systems*. **AAAI 2026** Demo.
- **Wu, Z., Cho, S.* et al. (2025).** *LibVulnWatch: A Deep Assessment Agent System and Leaderboard for Uncovering Hidden Vulnerabilities in Open-Source AI Libraries*. In Proceedings of the **ICML 2025** TAIG Workshop. <https://arxiv.org/abs/2505.08842> Demo: <https://huggingface.co/spaces/holistic-ai/LibVulnWatch>
- **Cho, S. et al. (2024).** *RTSUM: Relation Triple-based Interpretable Summarization with Multi-level Saliency Visualization*. In Proceedings of the **NAACL 2024** System Demonstrations Track. <https://aclanthology.org/2024.naacl-demo.5/>

EXPERIENCE

HOLISTIC AI

London, England, United Kingdom

Research Engineer, AI Startup

May 2025 – Present

- Reverse engineered the agent system's trace to build AI agent evaluation dataset with graph-based observability
- Won internal hackathon for a clustering algorithm by fine-tuning an embedding model with InfoNCE
- Designed trace-to-FSM reconstruction method achieving 0.975 AUROC failure detection and 68% compute savings
- Recently promoted to senior engineer after delivering a sandboxed agent system and real-time agent monitoring

KAKAO MOBILITY

Seoul, South Korea

Software Engineer, Digital Twin Team

December 2021 – September 2022

- Led a 3-person team to develop a vector-annotation platform integrated into the autonomous driving data pipeline
- Ported a C++ 3D-projection algorithm to Rust with Node.js bindings, making the mathematical computation library cross-platform

STRYX

Seoul, South Korea

Software Engineer, Geospatial AI Startup

November 2019 – December 2021

- Reduced build time by 70% and simplified dependency management by merging multiple repositories into a monorepo
- Downsized the Docker image by 90%, from 2GB to 180MB, by applying multi-stage builds in CI, elevating team productivity

EDUCATION

UNIVERSITY COLLEGE LONDON

London, England, United Kingdom

Artificial Intelligence for Sustainable Development MSc, Distinction

September 2024 – September 2025

- Dissertation: *Representation-Level Steering Improves Task Performance and Safety in LLMs*; Supervisor: Prof. Philip Treleaven; Mentor: Zekun Wu

YONSEI UNIVERSITY

Seoul, South Korea

Computer Science B.E.

March 2017 – August 2024

AWARDS

HERMES, 1ST PLACE (£3,000), HOLISTIC AI HACKATHON (2024)

London, England, United Kingdom

Team Lead

November 2024

- Fine-tuned SAE for GPT-2 to identify and steer correlated features for bias reduction in AI Safety
- Reduced stereotypical generation rate from 90% to 70% (22pp absolute) via SAE-derived steering vector

MBTIGPT, 1ST PLACE (₩3,000,000), YONSEI GENAI COMPETITION (2023)

Seoul, South Korea

Team Lead

September 2023 – November 2023

- Built AI service with 1,000+ users, RAG on chat history via MBTI analyzer using Redis and Faiss

PROJECTS

PaaT (Proprioception as a Tool)

March 2026 – Present

- Probe-as-tool framework for agent self-regulation; +11pt jailbreak / +18pt age-bias over baseline. ICML 2026 AIWILD Workshop under review.

OptimismBench

February 2026 – Present

- Built multilingual benchmark measuring directional bias in LLM probability judgment across 6 languages and 16+ models. NeurIPS 2026 Evaluations & Datasets Track target.

Stochastic Grokking

January 2026 – Present

- NeurIPS 2026 target. “A Head-Binding Lottery in Transformers.” Identifies head-binding lottery mechanism (27 to 52pp causal ablation gap vs normal heads); head transplantation rescues 6/6 failed seeds (mean 81% accuracy in 20k continued steps).

RouterRL

February 2026 – March 2026

- Trained only MoE router parameters (2.1M) via dense layer-wise probe rewards; matched SFT on ARC-Challenge (62.33% vs 62.00%).

TCPxAI

March 2024 – May 2024

- Built scalable TPCxAI big data pipelines on Hadoop and Spark cluster in GCP, supporting end-to-end AI workloads

ReSRer

September 2023 – January 2024

- Built pipeline indexing 21M documents into Milvus, accelerating LLM training across 4×RTX 3090 GPUs

LLaMa2GPTQ

June 2023 - July 2023

- Optimized memory cost by 75% using 4-bit GPTQ quantization applied to the LLaMa2 model and integrated RAG vector search

Texonom

November 2021 – June 2023

- Built an ANN-based vector retrieval API by embedding 30,000+ content pages into a Postgres pgvector with ONNX model

SKILLS

- **AI:** Python | PyTorch | AI agents | DDP/FSDP | PydanticAI | RAG | LangGraph | Agent SDK | Langfuse | LangSmith | ONNX
- **Service:** Full-Stack | TypeScript | Node.js | PostgreSQL | Redis | Rust | Vector Database | Faiss | Milvus | Vite | React | Next.js
- **Infra:** Kubernetes | Linux | Git | CI/CD | Docker | Apache Spark | Ansible | ETL | AWS | GCP | Hadoop | Terraform | SST

SERVICE

- Reviewer, ICML 2026
- Reviewer, ACL Rolling Review (March 2026)